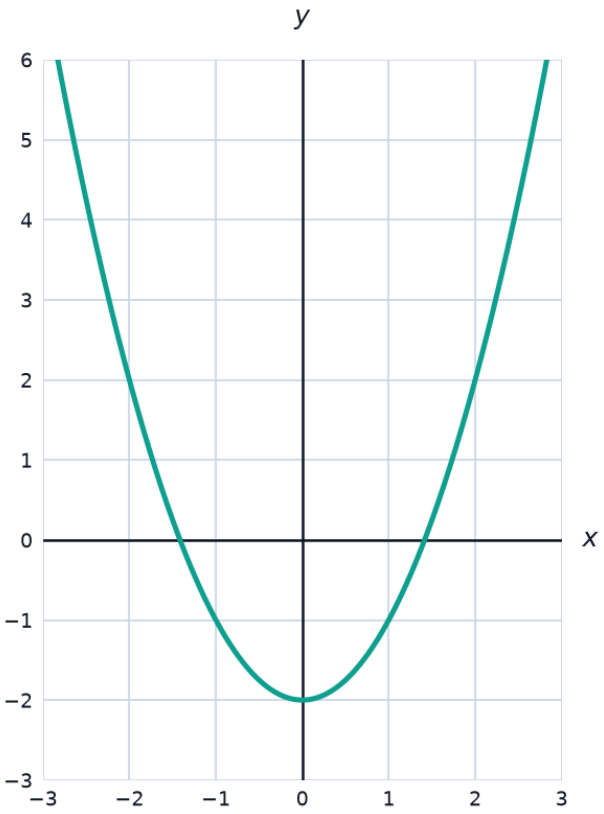
Functions and Function Notation

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Reading function graphs**

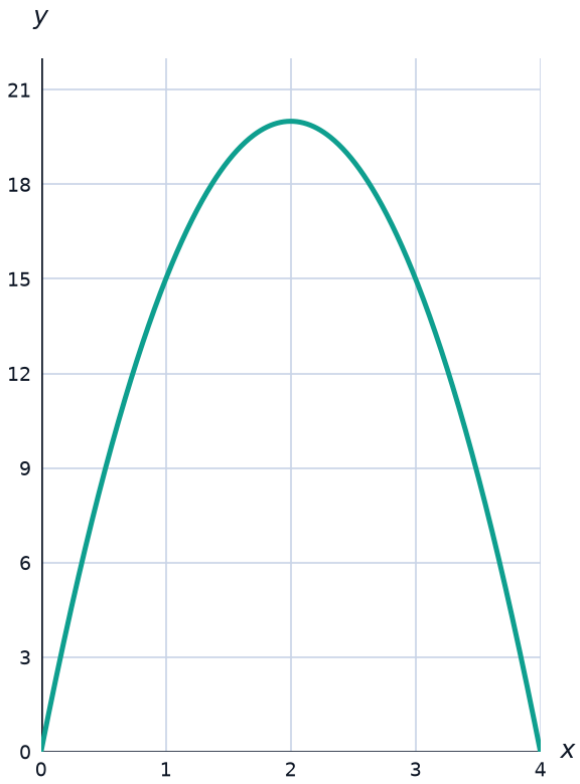
The graph of $f(x) = x^2 - 2$ is shown. Use it to estimate $f(1)$, and state the interval(s) where f is increasing.



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2 [Foundational] • Target: Average rate of change from function notation

A ball's height is modeled by $h(t) = -5t^2 + 20t$ (meters, seconds). Find the average rate of change of height from $t = 1$ to $t = 2$, and interpret the sign of your answer.



3 [Foundational] • Target: Function notation

Consider the function $f(x) = 2x^2 - 5x + 3$. Evaluate:

a $f(0)$ **b** $f(2)$ **c** $f(-1)$ **d** $f\left(\frac{1}{2}\right)$

4 [Foundational] • Target: Function notation

Consider the function $g(x) = \sqrt{x + 5}$. Evaluate:

a $g(4)$ **b** $g(-1)$ **c** $g(11)$ **d** $g(-5)$

5 [Foundational] • Target: Function notation

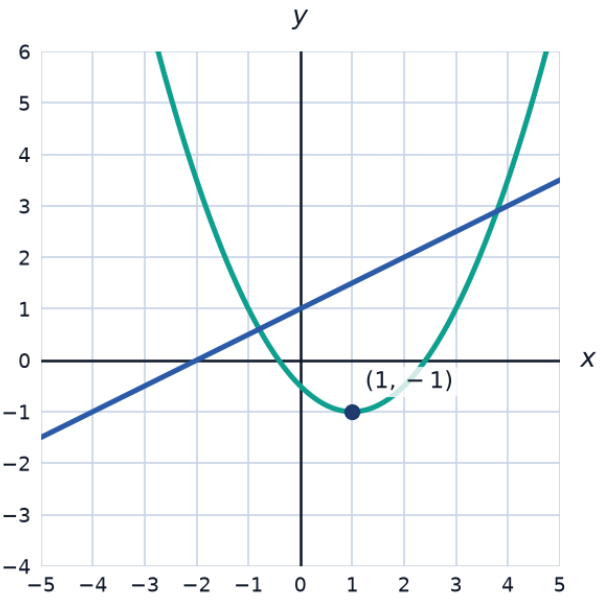
Consider the function $h(x) = \frac{x-3}{x+2}$. Evaluate:

a $h(0)$ **b** $h(3)$ **c** $h(-1)$ **d** $h(5)$

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6 [Foundational] • Target: Function notation

If $f(x) = 3x - 7$ and $f(a) = 11$, find the value of a.



7 [Foundational] • Target: Identifying functions

Determine whether each equation defines y as a function of x:

a $y = 2x + 1$

b $x = y^2$

c $y = |x|$

d $x^2 + y^2 = 9$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Identifying functions

Explain how the vertical line test is used to decide whether a graph represents a function.

9 [Intermediate] • Target: Combining functions

Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find:

a $(f + g)(x)$

b $(f - g)(x)$

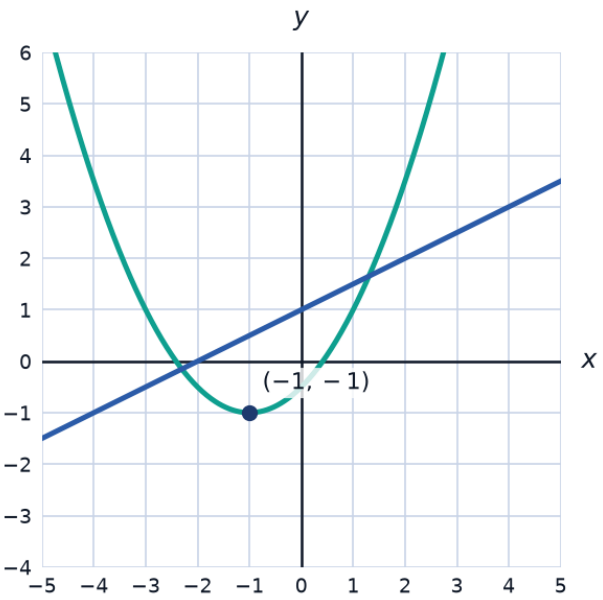
c $(f \cdot g)(2)$

d $(f \circ g)(1)$

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10 [Intermediate] • Target: Combining functions

A tutoring center charges a one-time registration fee plus an hourly rate, modeled by $C(n) = 250 + 45n$ riyals for n hours of tutoring.



a Interpret the value $C(0)$ in context.

b Find $C(12)$.

c Solve $C(n) = 925$ and interpret your answer.

11 [Intermediate] • Target: Combining functions

Let $f(x) = \sqrt{x}$ and $g(x) = x - 4$. Find the domain of $(f \circ g)(x)$.

12 [Intermediate] • Target: Piecewise-defined functions

A function is defined by $f(x) = x + 4$ for $x < 1$, and $f(x) = 2x + 1$ for $x \geq 1$. Evaluate:

a $f(-3)$ **b** $f(0)$ **c** $f(1)$ **d** $f(4)$

13 [Intermediate] • Target: Piecewise-defined functions

A phone plan charges a flat 60 riyals for up to 5 GB of data, then 15 riyals per additional GB beyond 5 GB.

a Write a piecewise formula $C(x)$ for the monthly cost, where x is data used in GB.

b Find $C(3)$ and $C(8)$.

c Is C continuous at $x = 5$? Explain.

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14 [Intermediate] • Target: Piecewise-defined functions

Sketch-reasoning: the graph of $f(x) = |x - 2|$ can be written as a piecewise function without absolute value bars. Write it as such, and state $f(-1)$ and $f(5)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Even, odd, and one-to-one functions • Cross-Lesson: 5.02 The Dot Product

Determine whether each function is even, odd, or neither:

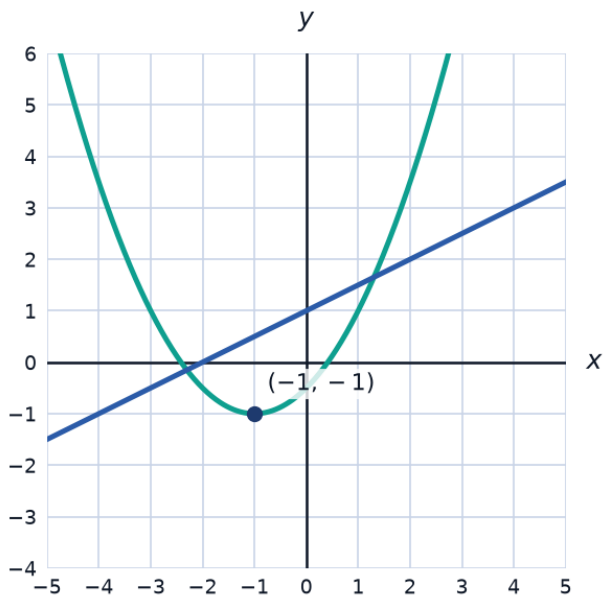
a $f(x) = x^4 - 3x^2$

b $f(x) = x^3 - x$

c $f(x) = x^2 + 2x$

16 [Challenging] • Target: Even, odd, and one-to-one functions • Cross-Lesson: 5.03 Parametric Equations

Determine whether each function is one-to-one (passes the horizontal line test). If not, give a horizontal line that crosses the graph twice.



a $f(x) = 2x + 5$

b $f(x) = x^2$

17 [Challenging] • Target: Reading function graphs • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Using the graph of $f(x) = x^2 - 2$ above, estimate all x-values where $f(x) = 2$.

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18 [Challenging] • Target: Average rate of change from function notation • Cross-Lesson: 6.02 Polar Equations and Graphs

For $f(x) = x^2 + 1$, find the average rate of change of f from $x = 1$ to $x = 3$.

19 [Challenging] • Target: Difference quotients • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

For $f(x) = x^2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$.

20 [Challenging] • Target: Difference quotients • Cross-Lesson: 7.02 Logarithms and Their Properties

For $f(x) = 3x + 2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$, and explain why the result makes sense for a linear function.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating Functions and Function Notation

Given $f(x) = 3x^2 - 4x + 1$, evaluate $f(-2)$.

a. 21

b. 5

c. -3

d. 17

22 [Foundational] • Target: Evaluating Functions and Function Notation

If $g(x) = \frac{2x + 5}{x - 1}$, find $g(3a + 1)$.

a. $\frac{6a + 7}{3a}$

b. $\frac{6a + 5}{3a + 1}$

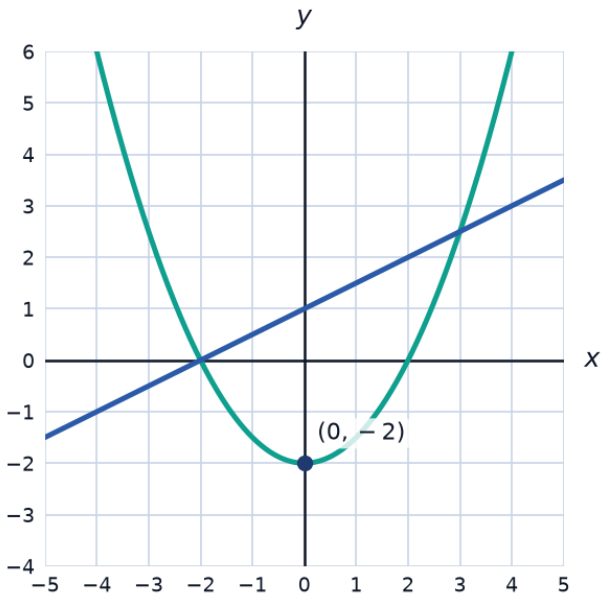
c. $\frac{6a + 2}{3a}$

d. $\frac{2a + 5}{a}$

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23 [Foundational] • Target: Evaluating Functions and Function Notation

Find all values of x for which $f(x) = x^2 - 6x$ equals -8.



a. $x = 2$ and $x = 4$

b. $x = -2$ and $x = -4$

c. $x = 1$ and $x = 8$

d. $x = 0$ and $x = 6$

24 [Foundational] • Target: The Vertical Line Test and Relations

Which of the following equations defines y as a function of x?

a. $y = 2x^3 - 5$

b. $x^2 + y^2 = 16$

c. $x = y^2 + 1$

d. $|y| = 3x$

25 [Foundational] • Target: The Vertical Line Test and Relations

What does the Vertical Line Test determine about a graph?

a. Whether the graph represents a function

b. Whether the function is one-to-one

c. The domain of the function

d. The slope of the curve

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26 [Foundational] • Target: The Vertical Line Test and Relations

Which set of ordered pairs represents a function?

a. $\{(1, 2), (2, 3), (3, 4), (4, 4)\}$

b. $\{(1, 2), (1, 3), (2, 4)\}$

c. $\{(0, 5), (0, -5), (1, 2)\}$

d. $\{(3, 1), (3, 2), (3, 3)\}$

27 [Foundational] • Target: Operations and Difference Quotients

Simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$ for $f(x) = 4x - 7$.

a. 4

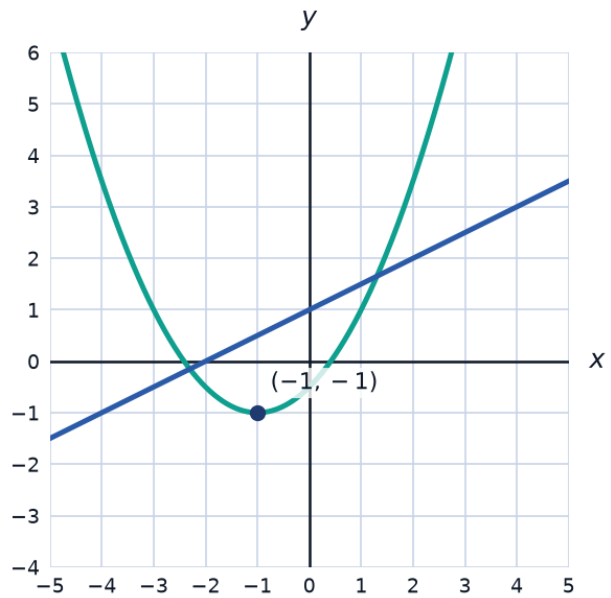
b. $4h$

c. $4x + 4h - 7$

d. 0

28 [Foundational] • Target: Operations and Difference Quotients

Simplify $\frac{f(x+h)-f(x)}{h}$ for $f(x) = 2x^2 + 3$.



a. $4x + 2h$

b. $4x$

c. $2x + 2h$

d. $4x + h$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Operations and Difference Quotients**

If $f(x) = x^2 - 1$ and $g(x) = 2x + 3$, find $(f \cdot g)(2)$.

a. 21

b. 15

c. 18

d. 28

30 **[Intermediate]** • **Target: Function Composition**

Given $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $(f \circ g)(x)$.

a. $2x^2 - 5$

b. $4x^2 + 4x - 2$

c. $2x^2 - 3$

d. $2x^2 + 1$

31 **[Intermediate]** • **Target: Function Composition**

For the same functions $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, evaluate $(g \circ f)(2)$.

a. 22

b. 25

c. 17

d. 3

32 **[Intermediate]** • **Target: Function Composition**

If $h(x) = \sqrt{2x^3 - 5}$, find functions $f(x)$ and $g(x)$ such that $h(x) = (f \circ g)(x)$.

a. $f(x) = \sqrt{x}, g(x) = 2x^3 - 5$

b. $f(x) = 2x^3 - 5, g(x) = \sqrt{x}$

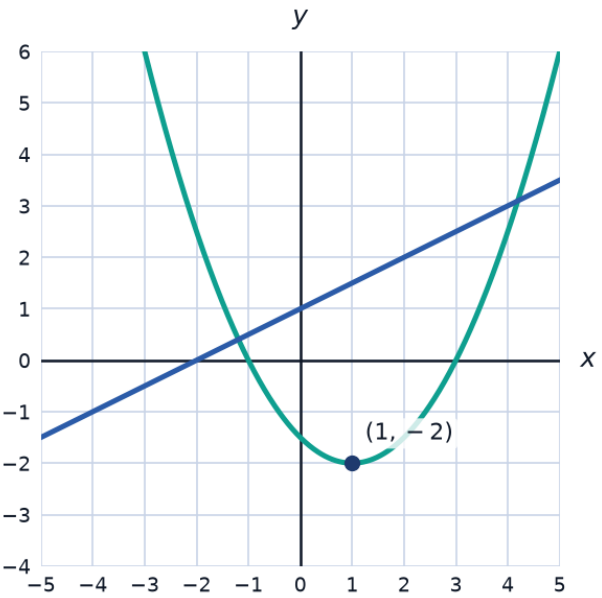
c. $f(x) = x^3, g(x) = \sqrt{2x - 5}$

d. $f(x) = 2\sqrt{x}, g(x) = x^3 - 5$

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33 [Intermediate] • Target: Function Composition

A function is defined by $f(x) = x^2$ for $x < 3$ and $f(x) = -2x + 1$ for $x \geq 3$. Find $f(3)$.



a. -6

b. -4

c. 9

d. -5

34 [Intermediate] • Target: Function notation

A function is defined by $f(x) = x^2$ for $x < 1$ and $f(x) = 2x + 1$ for $x \geq 1$. Find $f(1)$.

a. 4

b. 1

c. 3

d. 2

35 [Intermediate] • Target: Identifying functions

Determine whether $f(x) = 2x^4 - 5x^2 + 3$ is even, odd, or neither.

a. Even

b. Odd

c. Neither

d. Cannot be determined

36 [Intermediate] • Target: Combining functions

Determine whether $f(x) = x^5 + 2x$ is even, odd, or neither.

a. Even

b. Odd

c. Neither

d. Cannot be determined

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37 [Intermediate] • Target: Piecewise-defined functions

Determine whether $f(x) = x^5 - x^3$ is even, odd, or neither.

a. Even

b. Odd

c. Neither

d. Cannot be determined

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Even, odd, and one-to-one functions • Cross-Lesson: 6.02 Polar Equations and Graphs

A piecewise function is defined by $f(x) = 2x - 2$ for $x < -1$ and $f(x) = 3x + k$ for $x \geq -1$. Find k so that f is continuous at $x = -1$.

a. -1

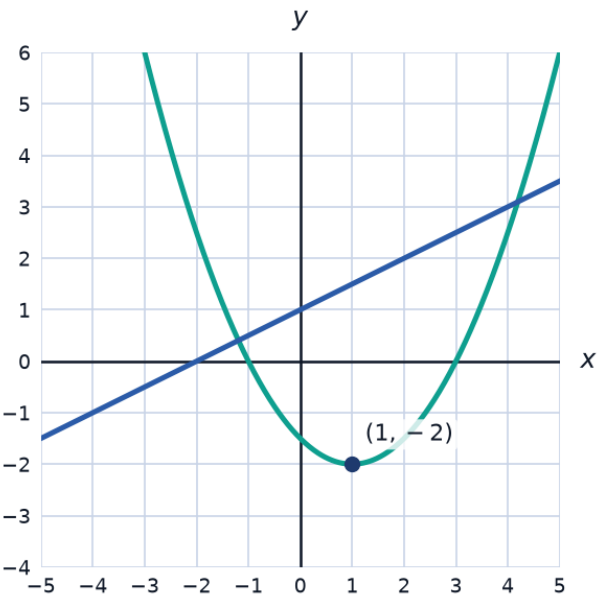
b. -4

c. -3

d. 1

39 [Challenging] • Target: Reading function graphs • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A piecewise function is defined by $f(x) = -2x + 3$ for $x < -1$ and $f(x) = 4x + k$ for $x \geq -1$. Find k so that f is continuous at $x = -1$.



a. 11

b. -4

c. 9

d. 5

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40 [Challenging] • Target: Average rate of change from function notation • Cross-Lesson: 7.02 Logarithms and Their Properties

A piecewise function is defined by $f(x) = x + 6$ for $x < -1$ and $f(x) = -2x + k$ for $x \geq -1$. Find k so that f is continuous at $x = -1$.

a. 5

b. 2

c. 4

d. 3

41 [Challenging] • Target: Difference quotients • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A piecewise function is defined by $f(x) = -x + 5$ for $x < 2$ and $f(x) = 2x + k$ for $x \geq 2$. Find k so that f is continuous at $x = 2$.

a. 4

b. 3

c. -1

d. 1

42 [Challenging] • Target: Inverse functions (introduction) • Cross-Lesson: 7.04 Applications: Growth and Decay

A piecewise function is defined by $f(x) = 2x + 4$ for $x < -2$ and $f(x) = x + k$ for $x \geq -2$. Find k so that f is continuous at $x = -2$.

a. 2

b. 4

c. -2

d. 0

43 [Challenging] • Target: Evaluating Functions and Function Notation • Cross-Lesson: 8.01 Rational Functions and Asymptotes

A piecewise function is defined by $f(x) = 2x + 1$ for $x < 3$ and $f(x) = -x + k$ for $x \geq 3$. Find k so that f is continuous at $x = 3$.

a. 7

b. 10

c. 12

d. -3

44 [Challenging] • Target: The Vertical Line Test and Relations • Cross-Lesson: 8.02 Graphing Rational Functions

A piecewise function is defined by $f(x) = 2x + 5$ for $x < -3$ and $f(x) = 3x + k$ for $x \geq -3$. Find k so that f is continuous at $x = -3$.

a. -9

b. 8

c. -1

d. 10

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45 [Challenging] • Target: Operations and Difference Quotients • Cross-Lesson: 8.03 Introduction to Limits

A piecewise function is defined by $f(x) = -3x + 1$ for $x < 2$ and $f(x) = x + k$ for $x \geq 2$. Find k so that f is continuous at $x = 2$.

a. 2

b. -5

c. -7

d. 3

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